

Machine learning



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Machine Learning

[Concepts, Methods]

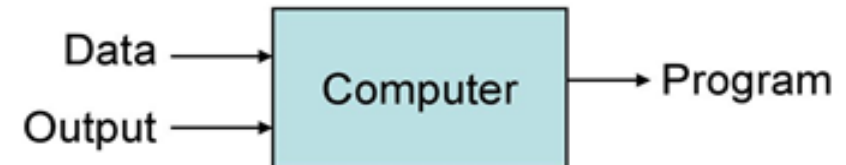
What is Machine Learning?

- Limitations of explicit programming
 - ✓ Spam filter: many rules
 - ✓ Automatic driving: too many rules
- Machine learning
 - ✓ "Field of study that gives computers the ability to learn without being explicitly programmed", Arthur Samuel (1959)

Traditional Programming



Machine Learning



What is Machine Learning?

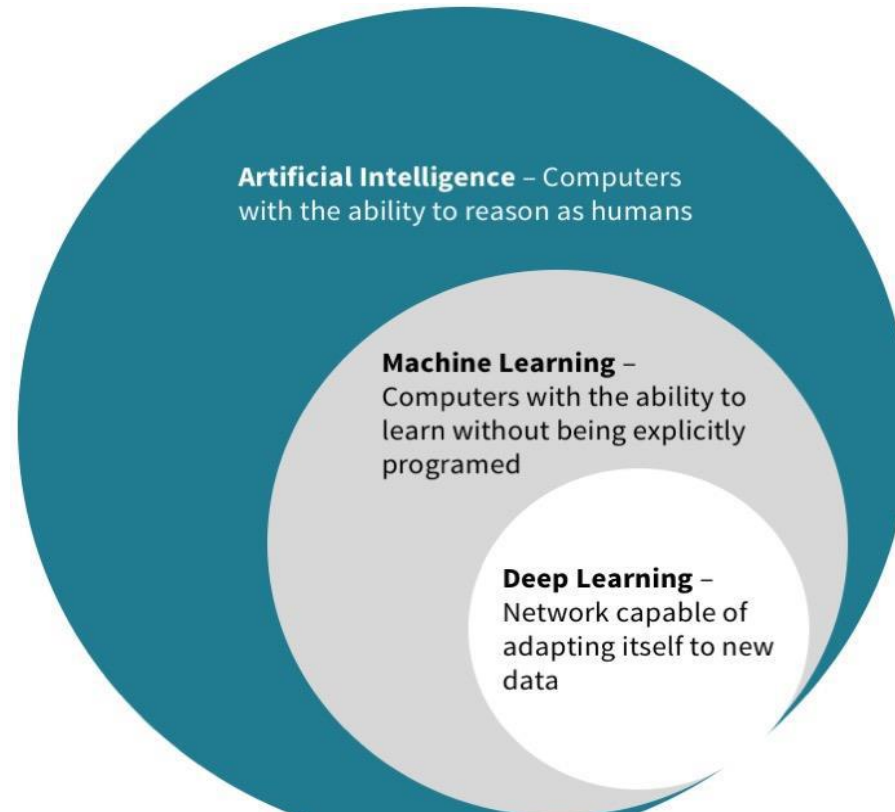
- Problems for which existing solutions require a lot of hand-tuning or longlists of rules
-> one machine learning algorithm can often simplify code and perform better
- Complex problems for which there is no good solution at all using a traditional approach -> the best machine learning techniques can find a solution
- Fluctuating environments -> a machine learning system can adapt to new data
- Getting insights about complex problems and large amounts of data

Terms often used interchangeably

- Data Mining

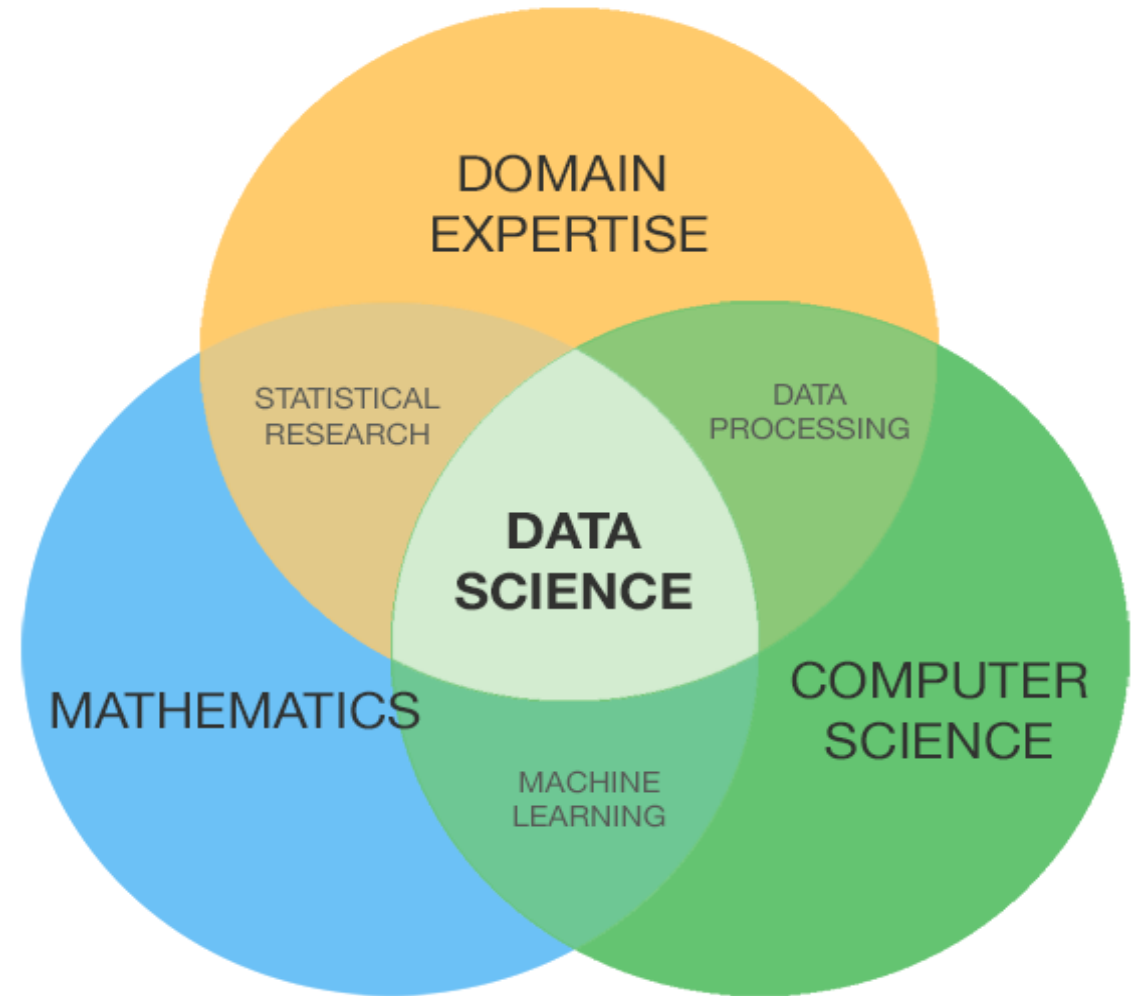
- ✓ Data analysis processes that apply ML techniques to solving real world problems

- AI & Deep Learning

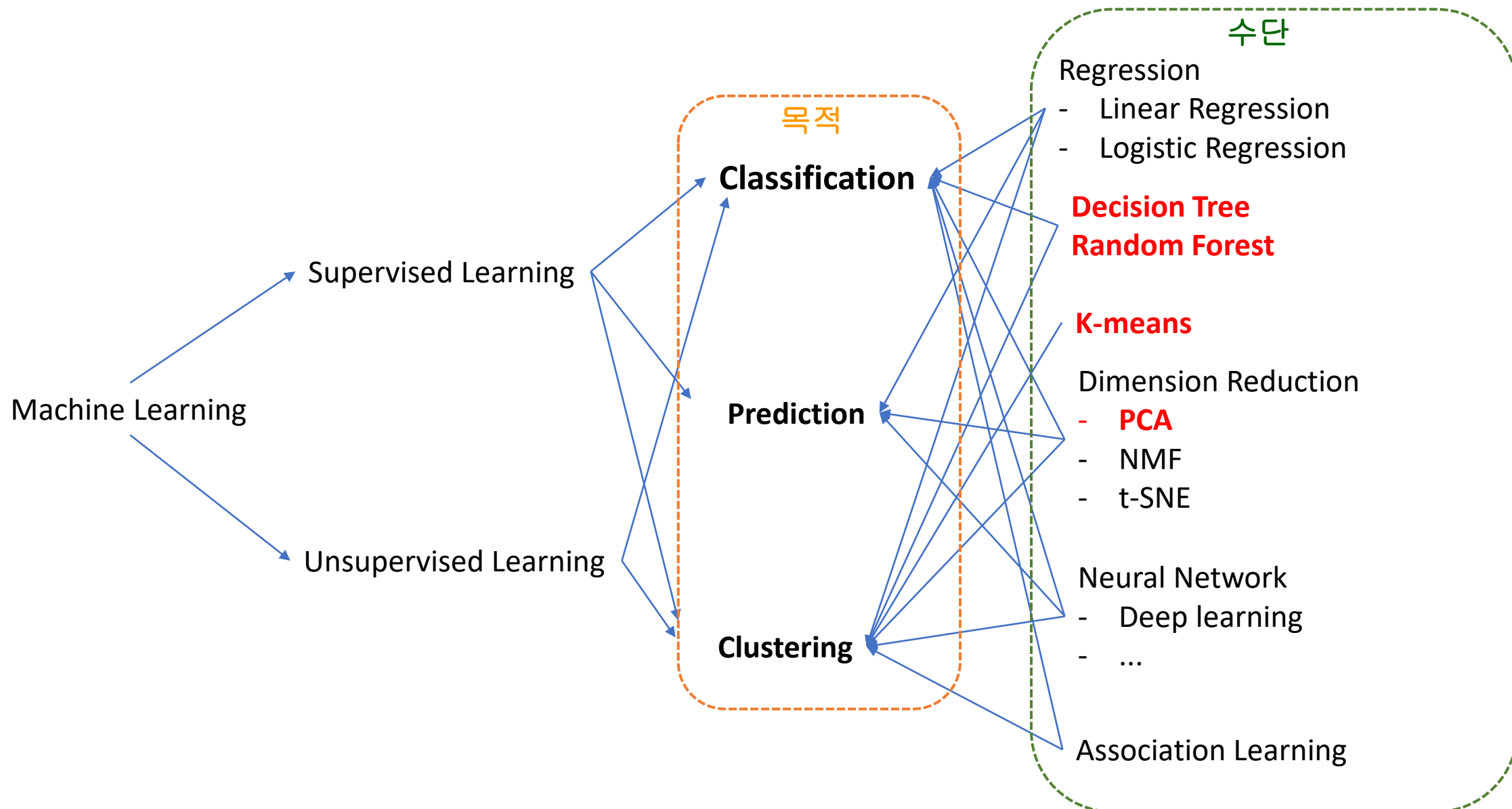


Statistical Analysis vs. Machine Learning

- Statistical Analysis
- Machine learning
- Data science
- Data scientist



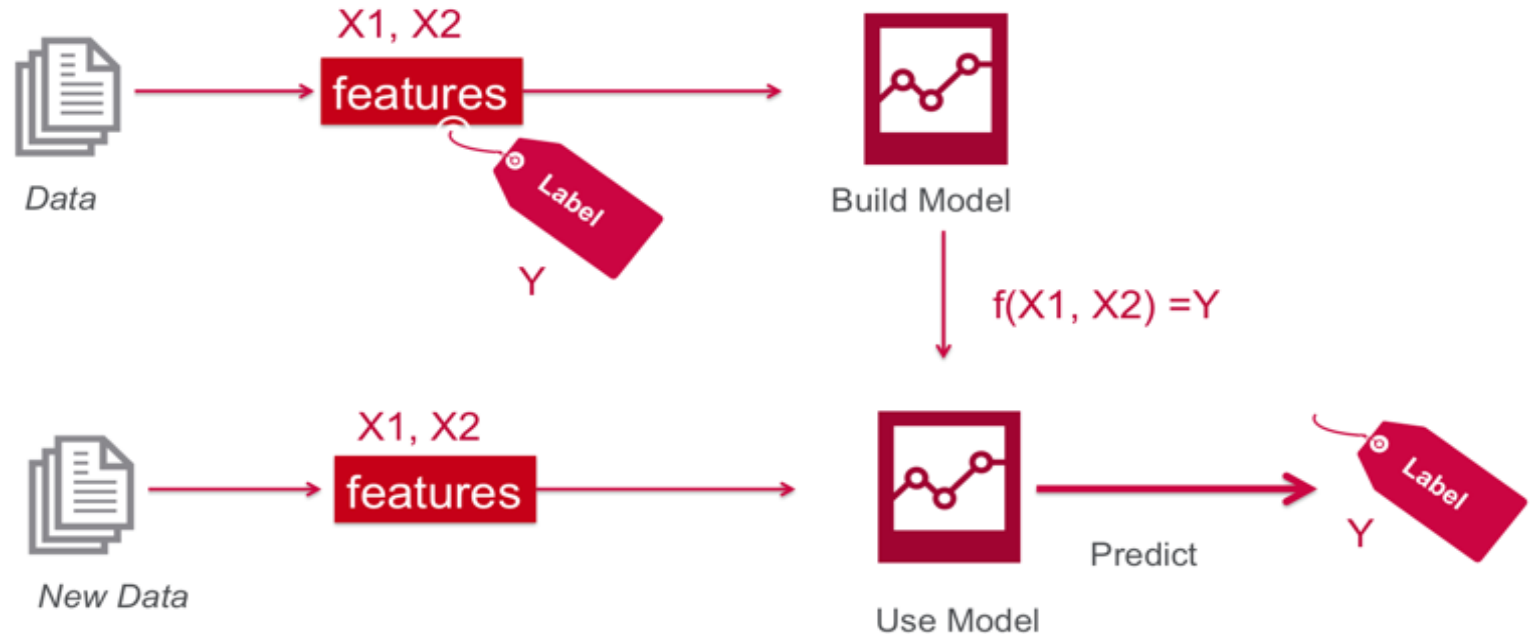
Types of Machine Learning



Supervised Learning

■ Algorithms

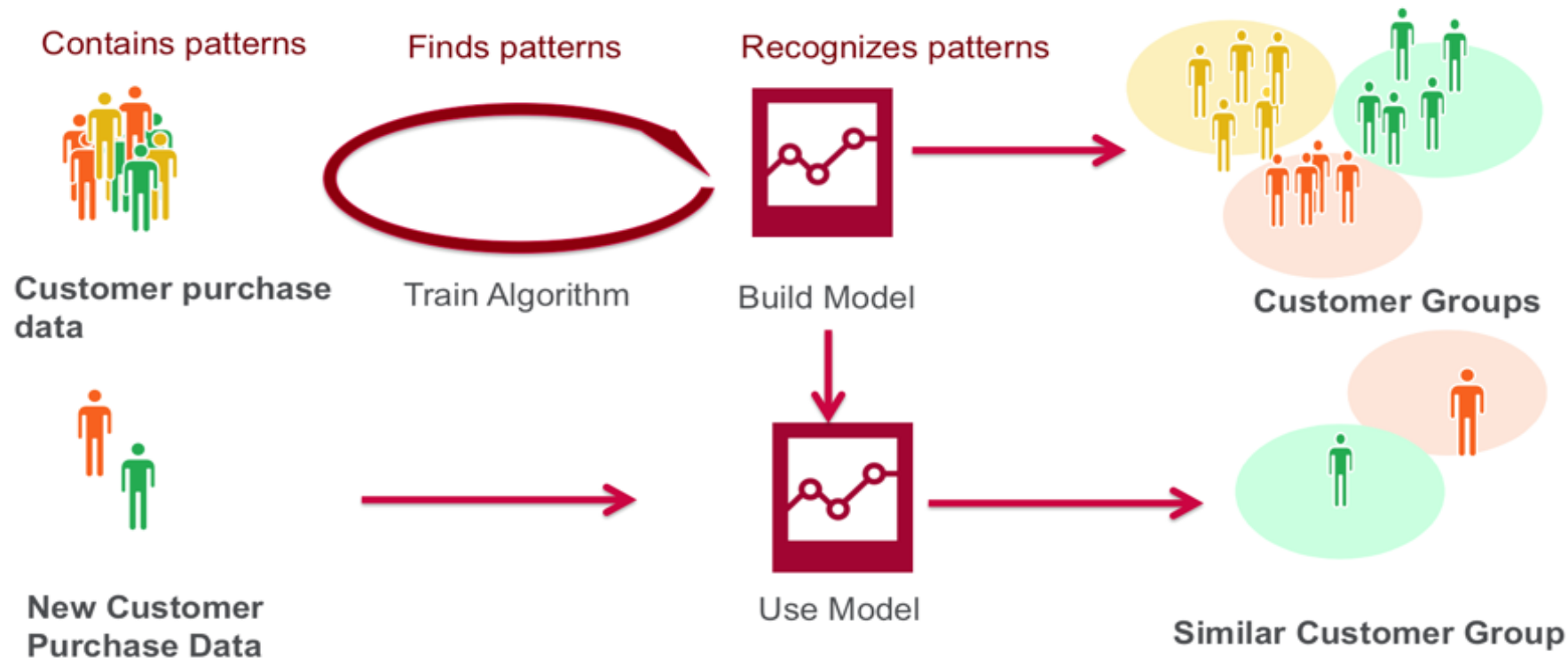
- ✓ K-Nearest Neighbor
- ✓ Linear Regression
- ✓ Logistic Regression
- ✓ Support Vector Machine (SVM)
- ✓ Decision Tree & Random Forest
- ✓ Neural Network



Unsupervised Learning

■ Algorithms

- ✓ Clustering:
 - ✓ K-Means, DBSCAN, EM, SOM
- ✓ Dimensionality reduction
 - ✓ PCA, NMF, t-SNE
- ✓ Association rule learning
 - ✓ Apriori, FP-growth



Question and Answer
THANKYOU 😊